

Undesirable response rate
(p_0)



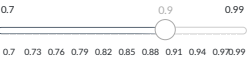
Acceptable response rate (p_1)



Type I error (α)



Power ($1 - \beta$)



Design Criterion

☒ Optimal ☐ MiniMax

Beta Prior on Response Rate

Beta(α , β) prior on p . Default Beta(1, 1) is uniform.

α (prior successes)

β (prior failures)

Simon Design

Posterior Distribution

Stage 1: enroll

17 patients

Stage 1 threshold (r_1)

≤ 3 responses $\rightarrow$ stop

Total sample size

37 patients

Final threshold (r)

> 10 responses $\rightarrow$
promising

Optimal Design — Summary Table

r_1 (Stage 1 threshold)	n_1 (Stage 1 sample)	r (Final threshold)	n (Total sample)	EN(p_0)	PET(p_0)
3.00	17.00	10.00	37.00	26.00	0.55

Decision Rules

Stage 1 stopping rule

Enroll 17 patients in Stage 1. If 3 or fewer responses are observed, stop the trial early and conclude that the treatment is not sufficiently active. Under the undesirable rate ($p_0 = 0.20$), there is a 54.9% probability of stopping early, with an expected trial size of only 26.0 patients.

Posterior probabilities at Stage 1 boundary — Beta(1.0, 1.0) prior, observing 3 / 17 responses:

$P(p \leq p_0 = 0.20 \mid \text{data}) = 0.499$ — posterior probability the response rate is undesirable or lower

$P(p \geq p_1 = 0.40 \mid \text{data}) = 0.033$ — posterior probability the response rate meets the acceptable target

Stage 2 decision rule

If more than 3 responses are seen in Stage 1, continue and enroll 20 additional patients in Stage 2 (total $N = 37$). At the end of the trial, if the total number of responses across both stages exceeds 10 out of 37 patients, declare the treatment promising. If 10 or fewer total responses are observed, conclude that the treatment does not meet the target activity level.

Posterior probabilities at final boundary — Beta(1.0, 1.0) prior, observing 10 / 37 responses:

$P(p \leq p_0 = 0.20 \mid \text{data}) = 0.122$ — posterior probability the response rate is undesirable or lower

$P(p \geq p_1 = 0.40 \mid \text{data}) = 0.057$ — posterior probability the response rate meets the acceptable target

'Optimal' minimises the expected sample size under p_0 (EN(p_0)).